Lab 3 Part 2 - Configuring AAA Authentication on Routers

Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway	Switch Port
R1	G0/1	192.168.1.1	255.255.255.0	N/A	S1 F0/1
	S0/0/0 (DCE)	10.1.1.2	255.255.255.252	N/A	N/A
R2	G0/0	192.168.2.1	255.255.255.0	N/A	S2 F0/2
	S0/0/0	10.1.1.1	255.255.255.252	N/A	N/A
	S0/0/1 (DCE)	10.2.2.1	255.255.255.252	N/A	N/A
R3	G0/1	192.168.3.1	255.255.255.0	N/A	S3 F0/5
	S0/0/1	10.2.2.2	255.255.255.252	N/A	N/A
TACACS+ Server	NIC	192.168.2.2	255.255.255.0	192.168.2.1	S2 F0/6
RADIUS Server	NIC	192.168.3.2	255.255.255.0	192.168.3.1	S3 F0/1
PC-A	NIC	192.168.1.3	255.255.255.0	192.168.1.1	S1 F0/2
PC-B	NIC	192.168.2.3	255.255.255.0	192.168.2.1	S2 F0/1
PC-C	NIC	192.168.3.3	255.255.255.0	192.168.3.1	S3 F0/18

Objectives

- Configure a local user account on R1 and configure authenticate on the console and vty lines using local AAA.
- Verify local AAA authentication from the R1 console and the PC-A client.
- Configure server-based AAA authentication using TACACS+.
- Verify server-based AAA authentication from the PC-B client.
- Configure server-based AAA authentication using RADIUS.
- Verify server-based AAA authentication from the PC-C client.

Background / Scenario

The network topology shows routers R1, R2 and R3. Currently, all administrative security is based on knowledge of the enable secret password. Your task is to configure and test local and server-based AAA solutions.

You will create a local user account and configure local AAA on router R1 to test the console and vty logins.

- User account: **Admin1** and password **admin1pa55**

You will then configure router R2 to support server-based authentication using the TACACS+ protocol. The TACACS+ server has been pre-configured with the following:

- Client: **R2** using the keyword **tacacspa55**
- User account: **Admin2** and password **admin2pa55**

Finally, you will configure router R3 to support server-based authentication using the RADIUS protocol. The RADIUS server has been pre-configured with the following:

- Client: **R3** using the keyword **radiuspa55**
- User account: **Admin3** and password **admin3pa55**

The routers have also been pre-configured with the following:

- Enable secret password: **ciscoenpa55**
- OSPF routing protocol with MD5 authentication using password: **MD5pa55**

Note: The console and vty lines have not been pre-configured.

Note: IOS version 15.3 supports SCRYPT as a secure encryption hashing algorithm; however, the IOS version that is currently supported in Packet Tracer uses MD5. Always use the most secure option available on your equipment.

Part 1: Configure Local AAA Authentication for Console Access on R1

Step 1: Test connectivity.

- Ping from **PC-A** to **PC-B**.
- Ping from **PC-A** to **PC-C**.
- Ping from **PC-B** to **PC-C**.

QUESTION 1 Which of the pings, if any, were successful?

Before proceeding, go to the "Lab 3 Part 2 - Configuring AAA Authentication on Routers - QUESTIONS" quiz on the Brightspace page and enter your answer for question 1. Leave the quiz open while you complete the rest of the lab sheet.

Step 2: Configure a local username on R1.

Configure a username of **Admin1** with a secret password of **admin1pa55**.

Step 3: Enable AAA on R1.

a. From global configuration mode on **R1**, enter the command **aaa authentication login TEST enable**.

```
R1(config)# aaa authentication login TEST enable
```

QUESTION 2 Was the command successful?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 2. Leave the quiz open while you complete the rest of the lab sheet.

b. Enable AAA on R1.

Step 4: Configure local AAA authentication for console access on R1.

Configure AAA authentication for logins to use the local database by default. Ensure the login entries are case sensitive.

Step 5: Configure the line console to use the defined AAA authentication method.

Configure the console login on R1 to use the default method list for authentication.

Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the local database. Try logging in using the wrong case in the username or password and note the results.

Part 2: Configure Local AAA Authentication for vty Lines on R1

Step 1: Configure domain name and crypto key for use with SSH.

a. Use netsecurity.com as the domain name on R1.

b. Create an RSA crypto key using 1024 bits.

Step 2: Configure a named list AAA authentication method for the vty lines on R1.

Configure a named list called **LOGIN-SSH** to authenticate logins using local AAA. The logins do not need to be case sensitive.

Step 3: Configure the vty lines to use the defined AAA authentication method.

Configure the vty lines to use the named AAA method and only allow SSH for remote access.

Step 4: Verify the AAA authentication method.

Verify the SSH configuration SSH to **R1** from the command prompt of **PC-A**.

QUESTION 3 Was PC-A able to successfully SSH to R1?

Before proceeding, return to the quiz on the Brightspace page and enter your answer for question 3. Leave the quiz open while you complete the rest of the lab sheet.

Part 3: Configure Server-Based AAA Authentication Using TACACS+ on R2

Step 1: Configure a backup local database entry called Admin.

For backup purposes, configure a local username of **Admin2** and a secret password of **admin2pa55**.

Step 2: Verify the TACACS+ Server configuration.

Click the TACACS+ Server. On the Services tab, click **AAA**. Notice that there is a Network configuration entry for **R2** and a User Setup entry for **Admin2**.

Step 3: Configure the TACACS+ server specifics on R2.

Configure the AAA TACACS server IP address and secret key on **R2**.

Note: The commands **tacacs-server host** and **tacacs-server key** are deprecated. Currently, Packet Tracer does not support the new command **tacacs server**.

```
R2(config)# tacacs-server host 192.168.2.2
R2(config)# tacacs-server key tacacspa55
```

Step 4: Configure AAA login authentication for console access on R2.

Enable AAA on **R2** and configure all logins to authenticate using the AAA TACACS+ server. If it is not available, then use the local database (not case sensitive).

Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA TACACS+ server.

Part 4: Configure Server-Based AAA Authentication Using RADIUS on R3

Step 1: Configure a backup local database entry called Admin.

For backup purposes, configure a local username of **Admin3** and a secret password of **admin3pa55**.

Step 2: Verify the RADIUS Server configuration.

Click the RADIUS Server. On the Services tab, click **AAA**. Notice that there is a Network configuration entry for **R3** and a User Setup entry for **Admin3**.

Step 3: Configure the RADIUS server specifics on R3.

Enter radius server configuration mode on **R3** using **RAD-SERVER** as the server name. Configure the AAA RADIUS server IP address of **192.168.3.2** and secret key of **radiuspa55**. (Do not change the authentication/accounting ports used as the server has already been pre-configured to use 1645.)

Note: Packet Tracer *does* support the new command **radius server**.

Step 4: Configure AAA login authentication for console access on R3.

Enable AAA on **R3** and configure all logins to authenticate using the AAA RADIUS server. If it is not available, then use the local database (case sensitive).

Step 5: Configure the line console to use the defined AAA authentication method.

Configure AAA authentication for console login to use the default AAA authentication method.

Step 6: Verify the AAA authentication method.

Verify the user EXEC login using the AAA RADIUS server.

If you have correctly configured all parts of the lab your activity score should now be showing as 100%. If so, click on "check results" in the activity window. Return to the Brightspace quiz one last time and enter the code into the appropriate question box (Q4) of the quiz.

You have completed the lab - please submit the Brightspace quiz.